

ACML101: Introduction to Chemistry
Inorganic Chemistry: Tutorial 1B

- (1) (a) How will you distinguish between cis and trans isomers of $[\text{CrCl}_2(\text{NH}_3)_4]\text{Cl}$?
(b) How will you determine whether MA_2B_2 complex is square planar or tetrahedral by an experiment?

(2) Propose an experiment to predict the structure of $[\text{MnBr}_4]^{2-}$. What are the limitations of VBT theory in predicting the properties of the compound. (How crystal field theory is useful in predicting few properties of metal complexes will be explained in the class)

3. Write down the d -orbital splitting diagram for $[\text{NiCl}_4]^{2-}$ and $[\text{Ni}(\text{CN})_4]^{2-}$. Will these complexes be paramagnetic or diamagnetic?

4. Δ_0 values of $[\text{Ru}(\text{H}_2\text{O})_6]^{2+}$ and $[\text{RuCl}_6]^{3-}$ are nearly same but according to spectrochemical series, the value for Δ_0 for the chlorine should be less compared to H_2O complex.

(5) How many geometric isomers are possible for $[\text{Pd}(\text{NH}_3)(\text{NMe}_3)\text{ClBr}]$? [why it has square planar geometry not tetrahedral, it will be discussed in the class]

Ans:

(6) How do you expect $(\text{C}_6\text{H}_5)_3\text{P}$, $(\text{C}_6\text{H}_5)_2\text{PO}$ and $(\text{C}_6\text{H}_5)_2\text{SO}$ to behave as coordinating ligands?

7. Calculate the CFSE values for $[\text{NiCl}_4]^{2-}$, $[\text{Mn}(\text{H}_2\text{O})_6]^{2+}$, $[\text{Co}(\text{NCS})_6]^{4-}$ and $[\text{Fe}(\text{CN})_6]^{4-}$. Explain why the CFSE for a low-spin d^6 configuration contains a 2P term? ((previous year question))

Orbital energy

8. Sketch the ionic radii curve for the first series of transition metal ions in M^{2+} oxidation state. Consider an octahedral geometry and cyanides as ligand. What will be the lowest point if the metal has +3 oxidation state?

(9) The magnetic moment of $[\text{CoL}_6]\text{SO}_4$ is 3.8 BM. Choose correct ligands from water, cyanide, chloride and ammonia, and draw the crystal-field splitting diagram. (previous year questions)

10. At room temperature, the observed value of μ_{eff} for $[\text{Cr}(\text{CH}_3\text{OCH}_2\text{CH}_2\text{OCH}_3)_3]\text{Br}_2$ is $4.75 \mu\text{B}$. Is the complex high-spin or low spin?

11. An aqueous solution of Co(III) sulphate is diamagnetic but becomes paramagnetic when a large excess of fluoride is added. Explain.

12. Solutions of $[\text{Cr}(\text{OH}_2)_6]^{3+}$ ions are pale blue green but the chromate ion, CrO_4^{2-} , is an intense yellow. Characterize the origins of the transitions and explain the relative intensities.

(13) The value of ϵ_{max} in the electronic spectra of $[\text{CuCl}_4]^{2-}$ and $[\text{Cu}(\text{OH}_2)_6]^{2+}$ differ by a factor of about 100. Comment on this observation and state which complex you expect to exhibit the larger value of ϵ_{max} . (previous year question)

14. Arrange the following species in the order of increasing Jahn Teller distortion in an octahedral environment high-spin cobalt (III) and low-spin Co(II)

15. Which complex is more stable $\text{cis}[\text{Cu}(\text{en})_2 2\text{H}_2\text{O}]$ or $\text{trans}[\text{Cu}(\text{en})_2 2\text{H}_2\text{O}]$